

OSSP (Operating Systems And Systems Programming) - 25CS2104E

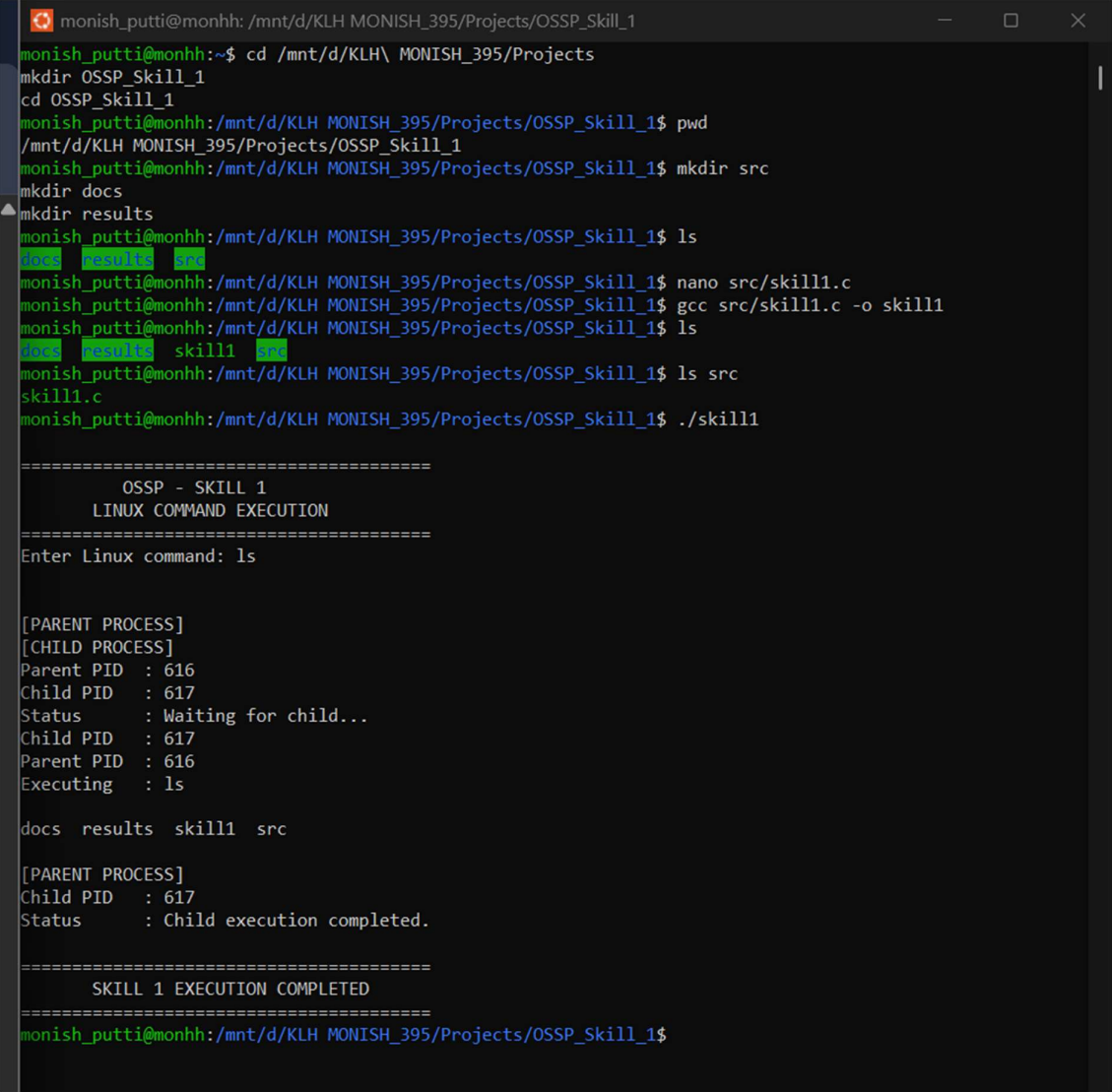
(SKILL Content Delivery Concepts)

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1. To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile

To Analyze process abstraction, execute fork(), understand exec() family, analyze parent-child relationships, inspect process tree, practice system call tracing



```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_1
monish_putti@monhh:~$ cd /mnt/d/KLH MONISH_395/Projects
mkdir OSSP_Skill_1
cd OSSP_Skill_1
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_1$ pwd
/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_1
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_1$ mkdir src
mkdir docs
mkdir results
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_1$ ls
docs  results  src
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_1$ nano src/skill1.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_1$ gcc src/skill1.c -o skill1
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_1$ ls
docs  results  skill1  src
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_1$ ls src
skill1.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_1$ ./skill1

=====
      OSSP - SKILL 1
      LINUX COMMAND EXECUTION
=====
Enter Linux command: ls

[PARENT PROCESS]
[CHILD PROCESS]
Parent PID   : 616
Child PID    : 617
Status       : Waiting for child...
Child PID    : 617
Parent PID   : 616
Executing    : ls

docs results skill1 src

[PARENT PROCESS]
Child PID    : 617
Status       : Child execution completed.

=====
      SKILL 1 EXECUTION COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_1$
```

2. To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_2
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_1$ mkdir ../OSSP_Skill_2
cd ../OSSP_Skill_2
mkdir src
mkdir docs
mkdir results
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_2$ pwd
ls
/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_2
docs results src
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_2$ nano src/skill2.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_2$ gcc src/skill2.c -o skill2
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_2$ ./skill2

=====
      OSSP - SKILL 2
      INTERACTIVE SHELL LOOP
=====
Type 'exit' to quit.

OSSP-Shell> ls

[COMMAND INPUT]
Command : ls
Status : Command received successfully.

OSSP-Shell> pwd

[COMMAND INPUT]
Command : pwd
Status : Command received successfully.

OSSP-Shell> hello

[COMMAND INPUT]
Command : hello
Status : Command received successfully.

OSSP-Shell> test command

[COMMAND INPUT]
Command : test command
Status : Command received successfully.

OSSP-Shell> exist
```

3. To Apply Escape Sequences, Store Command History, Navigate Previous Commands, Navigate Next Commands, Update Input Buffer, Test Recall Functionality.

To Allocate Buffers Dynamically, Resize Arrays, Prevent Buffer Overflow, Manage Linked Lists, Release Memory Correctly, Verify with Valgrind.

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_3
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_2$ cd /mnt/d/KLH\ MONISH_395/Projects
mkdir OSSP_Skill_3
cd OSSP_Skill_3
mkdir src
mkdir docs
mkdir results
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_3$ pwd
/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_3
ls
docs results src
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_3$ nano src/skill3.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_3$ gcc src/skill3.c -o skill3
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_3$ ./skill3

=====
                OSSP - SKILL 3
            COMMAND HISTORY & MEMORY
=====

Enter commands.
Type 'history' to display history.
Type 'exit' to finish.

OSSP-Shell> ls

[COMMAND STORED]
Command : ls
Status  : Stored in dynamic history list.

OSSP-Shell> pwd

[COMMAND STORED]
Command : pwd
Status  : Stored in dynamic history list.

OSSP-Shell> gcc

[COMMAND STORED]
Command : gcc
Status  : Stored in dynamic history list.

OSSP-Shell> history

----- COMMAND HISTORY -----
1. ls
2. pwd
3. gcc
-----

OSSP-Shell> exit

Exit command received.
```

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_3
docs result skill3 gcc
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_3$ valgrind --leak-check=full ./skill3
==827== Memcheck, a memory error detector
==827== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==827== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==827== Command: ./skill3
==827==

=====
OSSP - SKILL 3
COMMAND HISTORY & MEMORY
=====

Enter commands.
Type 'history' to display history.
Type 'exit' to finish.

OSSP-Shell> ls

[COMMAND STORED]
Command : ls
Status : Stored in dynamic history list.

OSSP-Shell>
```

4. To Split Input into Tokens, Identify Delimiters, Handle Whitespace, Create Token Structures, Validate Token Streams, Debug Parsing Output.

To Design Parser Logic, Generate Parse Trees, Validate Syntax, Detect Errors, Handle Empty Commands, Produce Execution Structures.

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_4
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_3$ cd /mnt/d/KLH\ MONISH_395/Projects
mkdir OSSP_Skill_4
cd OSSP_Skill_4
mkdir src
mkdir docs
mkdir results
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_4$ pwd
/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_4
ls
docs  results  src
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_4$ nano src/skill4.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_4$ gcc src/skill4.c -o skill4
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_4$ ./skill4

=====
O SSP - SKILL 4
COMMAND TOKENIZER & PARSER
=====

Enter a command: ls -1

[INPUT]
Command: ls

[TOKENIZATION]
Number of tokens: 2

----- TOKEN STREAM -----
Token 1 : ls
Token 2 : -1
-----

[VALIDATION]
Token stream is valid.

----- PARSE STRUCTURE -----
Command : ls
Arguments:
Argument 1 : -1
-----

[EXECUTION STRUCTURE]
Command ready for execution.

=====
SKILL 4 EXECUTION COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_4$
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_4$ ./skill4

=====
```

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_4

[EXECUTION STRUCTURE]
Command ready for execution.

=====
                SKILL 4 EXECUTION COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_4$
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_4$ ./skill4

=====
                OSSP - SKILL 4
                COMMAND TOKENIZER & PARSER
=====

Enter a command: ls      -l      src

[INPUT]
Command: ls

[TOKENIZATION]
Number of tokens: 3

----- TOKEN STREAM -----
Token 1 : ls
Token 2 : -l
Token 3 : src
-----

[VALIDATION]
Token stream is valid.

----- PARSE STRUCTURE -----
Command : ls
Arguments:
  Argument 1 : -l
  Argument 2 : src
-----

[EXECUTION STRUCTURE]
Command ready for execution.

=====
                SKILL 4 EXECUTION COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_4$
```

5. To Apply Single Quotes, Preserve Literal Content, Ignore Variable Expansion, Store Quoted Strings, Validate Parsing Results, Test Edge Cases.

To Apply Double Quotes, Preserve Spaces, Allow Variable Expansion, Parse Nested Tokens, Validate Outputs, Test Quoted Commands.

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_5
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_4$ cd /mnt/d/KLH\ MONISH_395/Projects
mkdir OSSP_Skill_5
cd OSSP_Skill_5
mkdir src
mkdir docs
mkdir results
pwd
/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_5
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_5$ nano src/skill5.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_5$ gcc src/skill5.c -o skill5
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_5$ ./skill5

=====
                OSSP - SKILL 5
                QUOTED STRING PARSER
=====

[SINGLE QUOTES]
Enter a single-quoted string: 'Hello World'

-----
                SINGLE QUOTE TEST
-----
Input      : 'Hello World'
Quoted content : 'Hello World'
Variable expansion : IGNORED
Literal content  : PRESERVED

[SINGLE QUOTE PARSING]
Token 1 : 'Hello
Token 2 : World'
Result : Single quotes validated successfully
Literal content preserved

[DOUBLE QUOTES]
Enter a double-quoted string: ./skill5

-----
                DOUBLE QUOTE TEST
-----
Input      : ./skill5
Quoted content : ./skill5
Spaces      : PRESERVED
Variable expansion : ALLOWED

[DOUBLE QUOTE PARSING]
Tokens detected : 1
Result : Double quotes validated successfully
Spaces inside quotes preserved

-----
```

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_5
Enter a single-quoted string: 'Hello World'

-----
SINGLE QUOTE TEST
-----
Input      : 'Hello World'
Quoted content : 'Hello World'
Variable expansion : IGNORED
Literal content  : PRESERVED

[SINGLE QUOTE PARSING]
Token 1 : 'Hello
Token 2 : World'
Result : Single quotes validated successfully
Literal content preserved

[DOUBLE QUOTES]
Enter a double-quoted string: ./skill5

-----
DOUBLE QUOTE TEST
-----
Input      : ./skill5
Quoted content : ./skill5
Spaces     : PRESERVED
Variable expansion : ALLOWED

[DOUBLE QUOTE PARSING]
Tokens detected : 1
Result : Double quotes validated successfully
Spaces inside quotes preserved

-----
EDGE CASE TEST
-----
Case 1 : Empty quoted string -> ""
Case 2 : Text containing spaces -> "hello world"
Case 3 : Literal variable -> '$HOME'
Case 4 : Expanded variable -> "$HOME"
Edge cases checked successfully.

=====
SKILL 5 EXECUTION COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_5$ '$HOME'
$HOME: command not found
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_5$
```

6. To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs.

To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching.

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_6
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_5$ cd /mnt/d/KLH\ MONISH_395/Projects
mkdir OSSP_Skill_6
cd OSSP_Skill_6
mkdir src
mkdir docs
mkdir results
pwd
/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_6
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_6$ nano src/skill6.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_6$ gcc src/skill6.c -o skill6
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_6$ ./skill6

=====
                OSSP - SKILL 6
            ESCAPE & PROCESS EXECUTION
=====

[PART 1: ESCAPE SEQUENCES]
Enter text containing an escape character: hello\ world

-----
                ESCAPE SEQUENCE TEST
-----
Input : hello\ world

[ESCAPE PROCESSING]
Character : h
Character : e
Character : l
Character : l
Character : o
Escaped character :
Character : w
Character : o
Character : r
Character : l
Character : d
Escape sequence processing completed.

[PART 2: CHILD PROCESS]
Enter a command to launch: ls

-----
                CHILD PROCESS EXECUTION
-----

[PARENT PROCESS]
PID      : 1132
Child PID : 1136
Status   : Waiting for child...

[CHILD PROCESS]
```

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_6

[CHILD PROCESS]
PID : 1136
PPID : 1132
Command : ls

Executing command...
docs results skill6 src

Child process completed.
Exit status : 0

=====
                SKILL 6 EXECUTION COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_6$ ./skill6

=====
                OSSP - SKILL 6
                ESCAPE & PROCESS EXECUTION
=====

[PART 1: ESCAPE SEQUENCES]
Enter text containing an escape character: file\ name.txt

-----
                ESCAPE SEQUENCE TEST
-----
Input : file\ name.txt

[ESCAPE PROCESSING]
Character : f
Character : i
Character : l
Character : e
Escaped character :
Character : n
Character : a
Character : m
Character : e
Character : .
Character : t
Character : x
Character : t
Escape sequence processing completed.

[PART 2: CHILD PROCESS]
Enter a command to launch:
```

7. Process Synchronization and Child Process Monitoring Using waitpid().

To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution.

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_7
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_6$ cd /mnt/d/KLH\ MONISH_395/Projects
mkdir OSSP_Skill_7
cd OSSP_Skill_7
mkdir src
mkdir docs
mkdir results
pwd
/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_7
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_7$ nano src/skill7.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_7$ gcc src/skill7.c -o skill7
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_7$ ./skill7

=====
                OSSP - SKILL 7
        PROCESS & COMMAND MONITOR
=====

=====
                PROCESS SYNCHRONIZATION
=====

[PARENT PROCESS]
PID       : 1230
Child PID : 1231
Status    : Waiting for child...

[CHILD PROCESS]
PID       : 1231
PPID      : 1230
Status    : Child is executing...
Child task completed.

[WAITPID RESULT]
Child process completed normally.
Exit status : 0
Parent continues after child completion.

Enter command to resolve: ls

=====
                COMMAND RESOLUTION
=====

[PATH VARIABLE]
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/Eclipse Adoptium/jdk-25.0.4.7-hotspot/bin:/mnt/c/Program Files/Eclipse Adoptium/jdk-25.0.4.7-hotspot/bin:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/windows/system32:/mnt/c/windows:/mnt/c/windows/System32/Wbem:/mnt/c/windows/System32/WindowsPowerShell/v1.0:/mnt/c/windows/System32/OpenSSH:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/c/WINDOWS/System32/OpenSSH:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/c/WINDOWS/System32/OpenSSH:/mnt/c/Program Files/nodejs:/
```

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_7
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_7$ ./skill7

=====
OSSP - SKILL 7
PROCESS & COMMAND MONITOR
=====

PROCESS SYNCHRONIZATION
=====

[PARENT PROCESS]
PID      : 1243
Child PID : 1244
Status   : Waiting for child...

[CHILD PROCESS]
PID      : 1244
PPID     : 1243
Status   : Child is executing...
Child task completed.

[WAITPID RESULT]
Child process completed normally.
Exit status : 0
Parent continues after child completion.

Enter command to resolve: gcc

=====
COMMAND RESOLUTION
=====

[PATH VARIABLE]
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/li
b:/mnt/c/Program Files/Eclipse Adoptium/jdk-25.0.4.7-hotspot/bin:/mnt/c/Program Files/Eclipse Adoptium/j
dk-25.0.4.7-hotspot/bin:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Fil
es (x86)/Common Files/Oracle/Java/javapath:/mnt/c/windows/system32:/mnt/c/windows:/mnt/c/windows/System3
2/Wbem:/mnt/c/windows/System32/WindowsPowerShell/v1.0:/mnt/c/windows/System32/OpenSSH:/mnt/c/WINDOWS/s
ystem32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt
/c/WINDOWS/System32/OpenSSH:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/
WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/c/WINDOWS/System32/OpenSSH:/mnt/c/Program Files/nodejs:/
mnt/c/Program Files/Git/cmd:/mnt/c/Program Files (x86)/cloudflare:/mnt/c/Program Files/MySQL/MySQL She
ll 8.0/bin:/mnt/c/Users/MONISH PUTTI/AppData/Local/Programs/Python/Python310/Scripts:/mnt/c/Users/MONI
SH PUTTI/AppData/Local/Programs/Python/Python310:/mnt/c/Users/MONISH PUTTI/AppData/Local/Programs/Pytho
n/Python314/Scripts:/mnt/c/Users/MONISH PUTTI/AppData/Local/Programs/Python/Python314:/mnt/c/Users/MON
ISH PUTTI/AppData/Local/Microsoft/WindowsApps:/mnt/c/users/monish putti/appdata/roaming/python/python314
/scripts:/mnt/c/users/monish putti/.local/bin:/mnt/c/Users/MONISH PUTTI/AppData/Local/Muse Hub/lib:/mnt/
c/Users/MONISH PUTTI/AppData/Local/Programs/Antigravity/bin:/mnt/c/Users/MONISH PUTTI/AppData/Roaming/np
m:/mnt/c/Users/MONISH PUTTI/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/MONISH PUTTI/AppDa
ta/Local/Python/bin:/snap/bin
```

8. To Detect Variable References, Expand Values, Handle Undefined Variables, Update Tokens, Support Nested Variables, Test Expansion Logic.

To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic.

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_8
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_7$ cd /mnt/d/KLH\ MONISH_395/Projects
mkdir OSSP_Skill_8
cd OSSP_Skill_8
mkdir src
mkdir docs
mkdir results
pwd
/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_8
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_8$ nano src/skill8.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_8$ gcc src/skill8.c -o skill8
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_8$ ./skill8

=====
                OSSP - SKILL 8
            VARIABLE & BUILT-IN DISPATCH
=====

[PART 1: VARIABLE EXPANSION]
Enter a variable reference: $HOME

-----
                VARIABLE EXPANSION
-----
Input : $HOME
Variable : $HOME
Value   : /home/monish_putti
Status  : Variable expanded successfully.

[PART 2: BUILT-IN DISPATCH]
Enter a built-in command: pwd
Current directory : /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_8

Dispatch status : BUILT-IN EXECUTED

=====
                SKILL 8 EXECUTION COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_8$
```

9. To Change Directories, Validate Paths, Update Working Directory, Handle Errors, Maintain Previous Directory, Test Navigation.

To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic.

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_9
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_9$ cd /mnt/d/KLH\ MONISH_395/Projects
mkdir OSSP_Skill_9
cd OSSP_Skill_9
mkdir src
mkdir docs
mkdir results
pwd
/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_9
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_9$ nano src/skill9.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_9$ gcc src/skill9.c -o skill9
src/skill9.c:7:9: warning: 'MAX_INPUT' redefined
 7 | #define MAX_INPUT 500
    | ^~~~~~
In file included from /usr/include/x86_64-linux-gnu/bits/local_lim.h:38,
                 from /usr/include/x86_64-linux-gnu/bits/posix1_lim.h:161,
                 from /usr/include/limits.h:198,
                 from /usr/lib/gcc/x86_64-linux-gnu/15/include/limits.h:210,
                 from /usr/lib/gcc/x86_64-linux-gnu/15/include/syslimits.h:9,
                 from /usr/lib/gcc/x86_64-linux-gnu/15/include/limits.h:34,
                 from src/skill9.c:5:
/usr/include/linux/limits.h:11:9: note: this is the location of the previous definition
11 | #define MAX_INPUT      255      /* size of the type-ahead buffer */
    | ^~~~~~
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_9$ ./skill9

=====
                OSSP - SKILL 9
        DIRECTORY & BUILT-IN DISPATCH
=====

Initial directory : /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_9

[/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_9]$ pwd

[PWD]
Current directory : /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_9

[/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_9]$ cd /tmp

[CD]
Previous directory : /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_9
Current directory  : /tmp
Status            : Directory changed successfully.

[/tmp]$ pwd

[PWD]
Current directory : /tmp

[/tmp]$
```

10. To Retrieve Current Directory, Display Path, Process Exit Requests, Cleanup Resources, Save State, Verify Correct Behavior.

To Parse Export Syntax, Update Environment Variables, Validate Names, Handle Existing Variables, Export to Child Processes, Test Functionality.

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_10
monish_putti@monhh:~$ cd /mnt/d/KLH\ MONISH_395/Projects
mkdir OSSP_Skill_10
cd OSSP_Skill_10
pwd
mkdir: OSSP_Skill_10: File exists
/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_10
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_10$ gcc skill10.c -o skill10
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_10$ ./skill10
=====
                OSSP - SKILL 10
            MINI SHELL ENVIRONMENT DEMO
=====

Supported commands:
  pwd
  export NAME=value
  env NAME
  exit

OSSP-SKILL10:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_10$ pwd
[PWD] /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_10

OSSP-SKILL10:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_10$ export STUDENT=MONISH
[EXPORT] STUDENT=MONISH

OSSP-SKILL10:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_10$ printenv STUDENT
Command execution failed: No such file or directory

[EXIT]
Saving shell state...
Cleaning up resources...
Shell terminated successfully.

OSSP-SKILL10:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_10$
```

11. To Store Commands, Maintain History Buffer, Handle Capacity Limits, Retrieve Entries, Display History, Validate Consistency.

To Create Pipeline Structures, Store Commands, Maintain Execution Order, Connect Processes, Validate Pipeline Layout, Test Configurations.

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_11
OSSP-SKILL10:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_10$ ^C
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_10$ cd /mnt/d/KLH\ MONISH_395/Projects
mkdir OSSP_Skill_11
cd OSSP_Skill_11
pwd
/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_11
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_11$ nano skill11.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_11$ gcc skill11.c -o skill11
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_11$ ./skill11
=====
OSSP - SKILL 11
COMMAND HISTORY
=====

History capacity : 5 commands

OSSP-SKILL11$ ls
[STORED] Command added to history.

OSSP-SKILL11$ pwd
[STORED] Command added to history.

OSSP-SKILL11$ date
[STORED] Command added to history.

OSSP-SKILL11$ whoami
[STORED] Command added to history.

OSSP-SKILL11$ uname
[STORED] Command added to history.

OSSP-SKILL11$ history

===== COMMAND HISTORY =====
1  ls
2  pwd
3  date
4  whoami
5  uname
=====

OSSP-SKILL11$ _
```

12. To Create Pipes, Redirect Streams, Execute Connected Commands, Close Descriptors, Synchronize Processes, Test Data Flow.

To Support Multiple Pipes, Manage Descriptor Arrays, Launch Multiple Processes, Coordinate Execution, Cleanup Resources, Test Long Pipelines.

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_12
3 date
4 whoami
5 uname
=====
OSSP-SKILL11$ ^C
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_11$ cd /mnt/d/KLH MONISH_395/Projects
mkdir OSSP_Skill_12
cd OSSP_Skill_12
pwd
/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_12
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_12$ nano skill12.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_12$ gcc skill12.c -o skill12
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_12$ ./skill12
=====
                OSSP - SKILL 12
                PIPE DEMO
=====

[PIPE] Pipe created successfully.

[PARENT PROCESS]
PID : 2076
Writing data to pipe...
Sent   : Hello from parent process!

[CHILD PROCESS]
PID : 2077
Reading data from pipe...
Received : Hello from parent process!
Child completed.

[SYNC] Parent waited for child.
[PIPE] Data flow completed successfully.

=====
                SKILL 12 DEMO COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_12$
```

13. To Parse Input Redirection, Open Files, Redirect stdin, Handle Missing Files, Restore Streams, Test File Inputs.

To Parse Output Redirection, Create Output Files, Redirect stdout, Handle Permissions, Verify Output Data, Test Redirection.

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_13
[SYNC] Parent waited for child.
[PIPE] Data flow completed successfully.

=====
SKILL 12 DEMO COMPLETED
=====

monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_12$ cd /mnt/d/KLH MONISH_395/Projects
mkdir OSSP_Skill_13
cd OSSP_Skill_13
pwd
/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_13
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_13$ nano skill13.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_13$ gcc skill13.c -o skill13
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_13$ ./skill13

=====
OSSP - SKILL 13
INPUT REDIRECTION
=====
input.txt: No such file or directory
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_13$ nano input.txt
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_13$ ls
input.txt skill13 skill13.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_13$ ./skill13

=====
OSSP - SKILL 13
INPUT REDIRECTION
=====

[REDIRECTION]
Standard input redirected from input.txt

[INPUT DATA]
Received : This data was read using input redirection.

[RESTORE]
Standard input restored successfully.

=====
SKILL 13 DEMO COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_13$
```

14. To Parse Append Operator, Open Files in Append Mode, Redirect Streams, Preserve Existing Data, Validate Appends, Test Scenarios.

To Parse stderr Redirection, Redirect Error Streams, Capture Error Output, Verify File Contents, Handle Failures, Test Error Cases.

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_14
SKILL 13 DEMO COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_13$ cd /mnt/d/KLH\ MONISH_395/Projects
mkdir OSSP_Skill_14
cd OSSP_Skill_14
pwd
/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_14
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_14$ nano skill14.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_14$ gcc skill14.c -o skill14
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_14$ echo "Original data in the file." > out
put.txt
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_14$ cat output.txt
Original data in the file.
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_14$ ./skill14
=====
OSSP - SKILL 14
APPEND DEMO
=====

[APPEND MODE]
File opened in append mode.
Data appended successfully.

[FILE]
Existing data was preserved.
New data was added at the end.

=====
SKILL 14 DEMO COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_14$ cat output.txt
Original data in the file.
New data appended to the file.
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_14$
```

15. To Parse Combined Redirection, Duplicate File Descriptors, Merge Streams, Verify Ordering, Handle Edge Cases, Test Integrated Outputs.

To Parse Complex Commands, Combine Pipes and Redirections, Resolve Execution Order, Validate Syntax, Generate Execution Plans, Test Scenarios.

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_15
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_14$ cat output.txt
Original data in the file.
New data appended to the file.
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_14$ cd /mnt/d/KLH\ MONISH_395/Projects
mkdir OSSP_Skill_15
cd OSSP_Skill_15
pwd
/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_15
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_15$ nano skill15.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_15$ gcc skill15.c -o skill15
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_15$ ./skill15
=====
OSSP - SKILL 15
COMBINED REDIRECTION
=====

[REDIRECTION]
Duplicating file descriptor...
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_15$ cat combined.txt
[STDOUT] Normal output redirected.
[STDERR] Error output redirected.
[RESULT] Both streams are merged into one file.
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_15$
```

16. To Detect Background Jobs, Launch Non-Blocking Processes, Return Prompt Immediately, Store Job Information, Monitor Execution, Test Background Tasks.

To Create Job Table, Store Process Groups, Track States, Update Status, Remove Completed Jobs, Validate Job Records.

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_16
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_16$ nano skill16.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_16$ gcc skill16.c -o skill16
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_16$ ./skill16
=====
OSSP - SKILL 16
BACKGROUND JOB DEMO
=====

[BACKGROUND JOB]
Launching a non-blocking process...

[PARENT]
Child PID : 2400
Status    : Background job started.
Prompt returned immediately.

[JOB INFORMATION]
Job PID   : 2400
Job State : RUNNING
[CHILD]
PID       : 2400
Status    : Background job running...
[CHILD]
PID       : 2400
Status    : Background job completed.

[JOB MONITOR]
Job PID   : 2400
Job State : COMPLETED

=====
SKILL 16 DEMO COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_16$
```

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_16

[JOB MONITOR]
Job PID : 2400
Job State : COMPLETED

=====
SKILL 16 DEMO COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_16$ nano skill16_jobs.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_16$ gcc skill16_jobs.c -o skill16_jobs
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_16$ ./skill16_jobs
=====
OSSP - SKILL 16
JOB TABLE
=====

[JOB TABLE]
Job ID : 1
PID : 2445
State : RUNNING

[PROCESS GROUP]
Process Group ID : 2444
Child PID : 2445

[CHILD PROCESS]
PID : 2445
Group : 2444
State : RUNNING

[JOB STATUS UPDATED]
Job ID : 1
PID : 2445
State : COMPLETED

[JOB TABLE]
Completed job removed from active jobs.

[VALIDATION]
Job record validated successfully.

=====
SKILL 16 DEMO COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_16$
```

17. To Implement how to List Active Jobs, Display Status Information, Retrieve Job IDs, Format Output, Update State Information, Test Job Listings.

To design how to Identify Target Job, Transfer Terminal Control, Resume Processes, Wait for Completion, Update Job State, Test Foreground Switching.

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_17
/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_17
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_17$ nano skill17.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_17$ gcc skill17.c -o skill17
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_17$ ./skill17
=====
O SSP - SKILL 17
ACTIVE JOBS
=====

[ACTIVE JOBS]
-----
Job ID    PID      State
-----
1         2507     RUNNING
-----

[JOB STATUS]
Job ID : 1
PID    : 2507
State  : RUNNING

[JOB ID]
Retrieved Job ID : 1

[UPDATED STATUS]
Job ID : 1
PID    : 2507
State  : COMPLETED

[JOB LIST]
No active jobs remaining.

=====
SKILL 17 DEMO COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_17$
```

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_17
=====
SKILL 17 DEMO COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_17$ nano skill17_foreground.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_17$ gcc skill17_foreground.c -o skill17_foreground
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_17$ ./skill17_foreground
=====
OSSP - SKILL 17
FOREGROUND SWITCHING
=====
[PARENT PROCESS]
Target Job PID : 2548
State          : WAITING

[TERMINAL CONTROL]
Parent waiting for foreground job...

[FOREGROUND JOB]
Target Job PID : 2548
State          : RUNNING

[TERMINAL CONTROL]
Foreground control transferred to child.

[PROCESS]
PID           : 2548
State         : COMPLETED

[JOB STATUS]
PID           : 2548
State         : COMPLETED

[FOREGROUND SWITCHING]
Foreground job completed.
Parent process resumed.

=====
SKILL 17 DEMO COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_17$
```

18. To Analyze Signal Delivery, Understand Signal Masks, Review Process Groups, Explore Terminal Signals, Test Signal Behavior.

To implement how to resume Stopped Jobs, Send Continue Signals, Update Job Status, Maintain Job Table, Verify Execution, Test Background Recovery.

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_18
pwd
/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_18
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_18$ nano skill18.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_18$ gcc skill18.c -o skill18
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_18$ ./skill18
=====
                OSSP - SKILL 18
                SIGNAL DELIVERY DEMO
=====

[SIGNAL SETUP]
SIGUSR1 handler registered.

[PARENT PROCESS]

[CHILD PROCESS]
PID      : 2633
PID      : 2634
Child PID: 2634
Group ID : 2633
Parent   : 2633
Group ID : 2633

[WAITING]
Child waiting for signal...

[SIGNAL DELIVERY]
Parent sending SIGUSR1 to child...

[SIGNAL HANDLER]
Signal received : 10
Signal delivered successfully.

[CHILD]
Signal received and processed.

[PROCESS GROUP]
Parent and child process relationship verified.

[SIGNAL TEST]
Signal delivery test completed successfully.

=====
                SKILL 18 DEMO COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_18$
```

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_18
Signal delivery test completed successfully.

=====
SKILL 18 DEMO COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_18$ nano skill18_resume.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_18$ gcc skill18_resume.c -o skill18_resume
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_18$ ./skill18_resume
=====
OSSP - SKILL 18
STOPPED JOB RECOVERY
=====

[JOB TABLE]
Job ID : 1
PID    : 2672
State  : RUNNING

[JOB]
PID    : 2672
State  : RUNNING

[JOB CONTROL]
Child stopping itself...

[JOB STATUS]
PID    : 2672
State  : STOPPED

[RECOVERY]
Sending SIGCONT to stopped job...
Continue signal sent successfully.

[JOB RESUMED]
PID    : 2672
State  : RUNNING AGAIN
Job completed successfully.

[JOB TABLE UPDATED]
Job ID : 1
PID    : 2672
State  : COMPLETED

[VALIDATION]
Stopped job resumed successfully.
Execution verified.

=====
SKILL 18 DEMO COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_18$
```

19. To Register Handlers, Configure Signal Actions, Handle Interrupts, Manage Signal Context, Verify Stability, Test Signal Events.

To Capture SIGINT, Forward Signals, Protect Shell Process, Terminate Foreground Jobs, Update Job States, Test Interrupt Handling.

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_19
pwd
/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_19
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_19$ nano skill19.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_19$ gcc skill19.c -o skill19
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_19$ ./skill19
=====
OSSP - SKILL 19
SIGNAL HANDLER DEMO
=====

[SIGNAL CONFIGURATION]
Signal : SIGUSR1
Handler registered successfully.

[PROCESS]
PID : 2735
Status : Waiting for signal...

[SIGNAL TEST]
Sending SIGUSR1 to current process...

[SIGNAL EVENT]
Signal received : 10
Handler executed successfully.

[STABILITY CHECK]
Process continued after signal handling.
Signal event handled successfully.

=====
SKILL 19 DEMO COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_19$
```

20. To Capture SIGTSTP, Suspend Processes, Update Job Table, Preserve State, Resume Later, Test Suspension Workflow.

To Create Process Groups, Assign Group IDs, Transfer Terminal Control, Restore Ownership, Coordinate Signals, Validate Behavior.

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_20
/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_20
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_20$ nano skill20.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_20$ gcc skill20.c -o skill20
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_20$ ./skill20
=====
OSSP - SKILL 20
PROCESS SUSPENSION DEMO
=====

[PARENT PROCESS]
PID      : 2819
Child PID: 2820

[CHILD PROCESS]
PID      : 2820
State    : RUNNING

[SUSPENSION]
Child requesting suspension...

[JOB TABLE]
Job ID   : 1
PID      : 2820
State    : STOPPED

[RESUME]
Sending SIGCONT to child...
Child resume signal sent.

[RESUMED]
PID      : 2820
State    : RUNNING AGAIN
Child completed execution.

[JOB TABLE UPDATED]
Job ID   : 1
PID      : 2820
State    : COMPLETED

[VALIDATION]
Suspension workflow verified.
Process state preserved and execution resumed.

=====
SKILL 20 DEMO COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_20$
```

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_20
SKILL 20 DEMO COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_20$ nano skill20_groups.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_20$ gcc skill20_groups.c -o skill20_groups
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_20$ ./skill20_groups
=====
OSSP - SKILL 20
PROCESS GROUP DEMO
=====

[PARENT PROCESS]
PID      : 2856
[CHILD PROCESS]
Child PID : 2857
PID       : 2857
Parent PID: 2856
Process Group ID : 2857
State     : GROUP CREATED

[SIGNAL COORDINATION]
Child waiting for parent signal...

[PROCESS GROUP]
Creating process group for child...
Process Group ID : 2857
Group created successfully.

[TERMINAL CONTROL]
Terminal ownership coordinated by parent.
Child process group is active.

[SIGNAL COORDINATION]
Sending SIGCONT to process group...
Signal sent to process group successfully.
Child received coordination request.
Child execution completed.

[RESTORE OWNERSHIP]
Parent process restored terminal ownership.

[VALIDATION]
Process group ID verified.
Signal coordination verified.
Process group execution completed successfully.

=====
SKILL 20 DEMO COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_20$
```

21. To Connect Modules, Validate Interfaces, Test End-to-End Execution, Handle Errors, Improve Reliability, Refactor Components.

To Detect Runtime Errors, Generate Messages, Log Failures, Handle Invalid Syntax, Recover Gracefully, Test Error Scenarios.

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_21
/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_21
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_21$ nano skill21.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_21$ gcc skill21.c -o skill21
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_21$ ./skill21
=====
O SSP - SKILL 21
MODULE INTEGRATION
=====

[MODULE CONNECTION]
Connecting parent and child modules...
Parent module connected successfully.

[CHILD MODULE]
Child module started successfully.
Child PID : 2924
Module execution completed.
Child PID : 2924

[INTERFACE VALIDATION]
Parent-to-child interface verified.
Process communication interface is valid.

[END-TO-END TEST]
Child module completed successfully.
End-to-end execution verified.

[RELIABILITY CHECK]
All connected modules completed normally.
System execution is reliable.

[REFACTOR CHECK]
Child functionality maintained as a separate module.
Module interface remains valid.

=====
SKILL 21 DEMO COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_21$
```

22. To Run Valgrind, Identify Leaks, Fix Memory Errors, Release Resources, Verify Cleanup, Document Findings.

To Design Test Cases, Create Test Scripts, Automate Execution, Verify Outputs, Compare Results, Generate Reports.

```

monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_22
pwd
/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_22
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_22$ nano skill22.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_22$ gcc skill22.c -o skill22
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_22$ ./skill22
=====
OSSP - SKILL 22
VALGRIND DEMO
=====

[MEMORY ALLOCATION]
Memory allocated successfully.
Stored values: 10 20 30 40 50

[MEMORY RELEASE]
Allocated memory released successfully.

[CLEANUP VERIFICATION]
Memory pointer cleared.
Cleanup verified successfully.

[RESULT]
No intentional memory leak remains.
Memory management completed successfully.

=====
SKILL 22 DEMO COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_22$ valgrind --leak-check=full ./skill22
==2997== Memcheck, a memory error detector
==2997== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==2997== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==2997== Command: ./skill22
==2997==
=====
OSSP - SKILL 22
VALGRIND DEMO
=====

[MEMORY ALLOCATION]
Memory allocated successfully.
Stored values: 10 20 30 40 50

[MEMORY RELEASE]
Allocated memory released successfully.

[CLEANUP VERIFICATION]
Memory pointer cleared.
Cleanup verified successfully.

[RESULT]
No intentional memory leak remains.

```

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_22
=====
VALGRIND DEMO
=====
[MEMORY ALLOCATION]
Memory allocated successfully.
Stored values: 10 20 30 40 50

[MEMORY RELEASE]
Allocated memory released successfully.

[CLEANUP VERIFICATION]
Memory pointer cleared.
Cleanup verified successfully.

[RESULT]
No intentional memory leak remains.
Memory management completed successfully.

=====
SKILL 22 DEMO COMPLETED
=====
==2997==
==2997== HEAP SUMMARY:
==2997==    in use at exit: 0 bytes in 0 blocks
==2997==   total heap usage: 2 allocs, 2 frees, 1,044 bytes allocated
==2997==
==2997== All heap blocks were freed -- no leaks are possible
==2997==
==2997== For lists of detected and suppressed errors, rerun with: -s
==2997== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_22$
```

23. To Execute Large Pipelines, Launch Multiple Jobs, Measure Stability, Monitor Resources, Detect Failures, Analyze Performance.

To Document Architecture, Create Flow Diagrams, Explain Execution Paths, Describe Syscalls, Record Design Decisions, Review Documentation.

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_23
/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_23
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_23$ nano skill23.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_23$ gcc skill23.c -o skill23
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_23$ ./skill23
=====
O SSP - SKILL 23
PIPELINE STRESS DEMO
=====

[PIPELINE]
Launching multiple pipeline jobs...
Number of jobs : 4

[CHILD JOB 1]
PID : 3063
Status : Running

[CHILD JOB 2]

[CHILD JOB 3]
PID : 3064
PID : 3065
Status : Running
Status : Running

[PARENT PROCESS]
All pipeline jobs launched.
Monitoring child processes...

[CHILD JOB 4]
PID : 3066
Status : Running
Job 1 completed successfully.
Job 3 completed successfully.
Job 2 completed successfully.
Job 4 completed successfully.
Child PID 3063 : SUCCESS
Child PID 3064 : SUCCESS
Child PID 3065 : SUCCESS
Child PID 3066 : SUCCESS

[STABILITY CHECK]
All pipeline jobs were monitored.
All child processes completed.

[RESOURCE MONITOR]
Parent PID : 3062
Jobs launched : 4
Jobs completed : 4

[PERFORMANCE]
Execution time : 1 seconds
```

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_23

[PARENT PROCESS]
All pipeline jobs launched.
Monitoring child processes...

[CHILD JOB 4]
PID      : 3066
Status   : Running
Job 1 completed successfully.
Job 3 completed successfully.
Job 2 completed successfully.
Job 4 completed successfully.
Child PID 3063 : SUCCESS
Child PID 3064 : SUCCESS
Child PID 3065 : SUCCESS
Child PID 3066 : SUCCESS

[STABILITY CHECK]
All pipeline jobs were monitored.
All child processes completed.

[RESOURCE MONITOR]
Parent PID      : 3062
Jobs launched   : 4
Jobs completed  : 4

[PERFORMANCE]
Execution time  : 1 seconds

[FAILURE CHECK]
No child process failure detected.

[ANALYSIS]
Multiple jobs executed successfully.
Pipeline workload completed normally.

=====
                SKILL 23 DEMO COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_23$
```

24. To Organize Source Code, Improve Build Scripts, Add README, Verify Repository Structure, Tag Releases, Validate Deliverables.

To Create Demo Scenarios, Validate Features, Rehearse Presentation, Demonstrate Pipelines, Demonstrate Signals, Final Project Review.

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_24
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_24$ nano skill24.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_24$ gcc skill24.c -o skill24
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_24$ ./skill24
=====
OSSP - SKILL 24
PROJECT REVIEW DEMO
=====

[SOURCE CODE]
Source code organization verified.

[BUILD SCRIPT]
Build configuration verified.

[README]
Project README is available.

[REPOSITORY]
Repository structure verified.

[DELIVERABLES]
Project files validated successfully.

[RELEASE]
Release structure prepared successfully.

=====
SKILL 24 DEMO COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_24$
```

```
monish_putti@monhh: /mnt/d/KLH MONISH_395/Projects/OSSP_Skill_24
SKILL 24 DEMO COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_24$ nano skill24_demo.c
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_24$ gcc skill24_demo.c -o skill24_demo
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_24$ ./skill24_demo
=====
OSSP - SKILL 24
FINAL PROJECT DEMO
=====

[DEMO SCENARIO]
Starting final project demonstration...

[FEATURE VALIDATION]
Parent PID : 3163
Child PID  : 3164
Pipe created successfully.

[PIPELINE DEMONSTRATION]
Message sent from parent to child.

[CHILD PROCESS]
PID : 3164
Received through pipeline:
Pipeline communication successful.

[SIGNAL]
Child sending SIGUSR1 to parent...

[SIGNAL DEMO]
SIGUSR1 received by process 3163.
Signal handling verified.

[PRESENTATION CHECK]
Pipeline feature demonstrated.
Signal feature demonstrated.
Parent-child execution verified.

[FINAL PROJECT REVIEW]
Process creation      : VERIFIED
Pipeline communication : VERIFIED
Signal handling       : VERIFIED
Process synchronization : VERIFIED
Final demonstration   : COMPLETED

=====
SKILL 24 DEMO COMPLETED
=====
monish_putti@monhh:/mnt/d/KLH MONISH_395/Projects/OSSP_Skill_24$
```